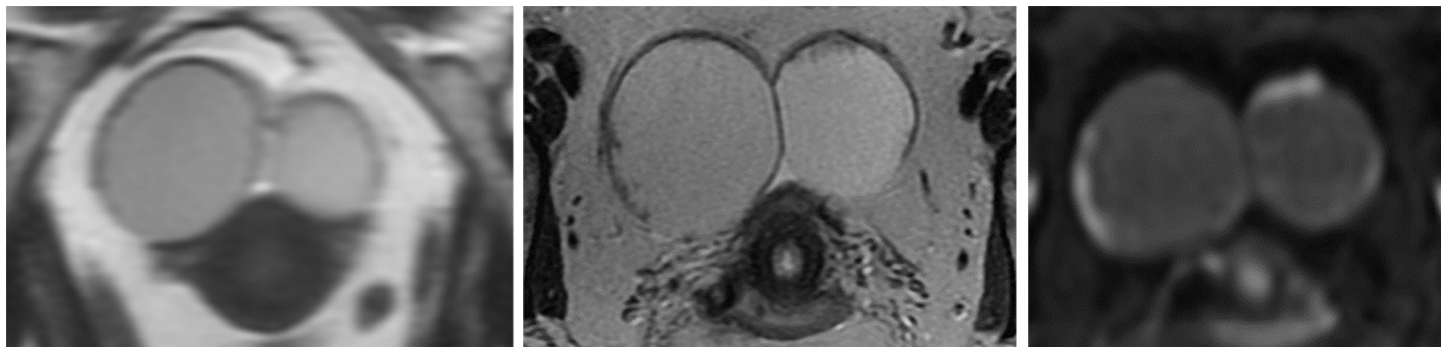


Optimizing the MRI Pelvis Protocol for O-RADS MRI

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See the invited commentary by [Lakhman and Rakow-Penner](#) in this issue.



Adnexal lesions seen on US images are referred for MRI when there is a need for further characterization and more accurate cancer risk score assessment. MRI can help expedite referral to gynecologic oncologists when ovarian cancers are suspected and avoid unnecessary, overextensive surgery or unnecessary follow-up when benign lesions are identified. The Ovarian-Adnexal Reporting and Data System (O-RADS) MRI was developed to provide standardized lexicon terminology and to allow predicting the risk of cancer when the appropriate MRI protocol is used. Only when the MRI protocol with the minimum technical requirements and dynamic contrast-enhanced imaging is used can the positive predictive values listed in the O-RADS MRI table be obtained. The O-RADS MRI recommended protocol includes multiplanar T2-weighted imaging, diffusion-weighted imaging, in- and opposed-phase imaging, and pre- and postcontrast imaging, including dynamic contrast-enhanced imaging. Optimizing each MRI sequence may be challenging but will lead to accurate risk stratification of adnexal lesions. It is also important to understand how each sequence is used to assess adnexal lesion characteristics for accurate risk scoring. This article reviews the necessary sequences in the recommended O-RADS MRI protocol and offers tips and tricks that can be used to optimize images, negate common artifacts, and avoid common errors in interpretation, including when the MRI examination is suboptimal.

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Introduction

Adnexal lesions are commonly encountered in clinical practice, and a combination of US and MRI are used to characterize them. US serves as the primary imaging modality with high sensitivity for the detection of malignancy. MRI is a problem-solving tool and has improved specificity for the diagnosis of malignancy due to its ability to detect enhancing solid tissue (1–9). When there is no enhancing solid tissue, MRI can exclude malignancy with a very high negative predictive value of 98% (5). Over 75% of adnexal lesions referred from US as potentially malignant are ultimately characterized as benign at MRI, which allows for conservative management and is reassuring to the patient (10). Furthermore, MRI has the potential of reducing the number of surgical evaluations of benign cysts by 90% and has been shown to decrease the referral rate to gynecologic oncologists by approximately 70% (10,11). When a lesion is found to be potentially malignant at MRI, the patient can be expeditiously referred to a gynecologic oncologist for improved patient outcomes (12,13).

To aid in the assessment of adnexal lesions and cancer risk assessment, the American College of Radiology Ovarian-Adnexal Reporting and Data System (O-RADS) MRI committee published an MRI-based lexicon with standardized terminology and a data-driven cancer risk score (14). This system has been shown in multiple studies to have sensitivities above 90% and specificities of 91%, with high negative predictive values greater than 98% and positive predictive values (PPVs) of 74% (9,15). The O-RADS MRI risk stratification system relies on MRI's capacity to accurately distinguish between fluid and solid tissue components and to assess whether the enhancing elements represent true solid tissue. To attain the listed PPV for malignancy seen in the O-RADS MRI Risk Score Assessment Table available on the American College of Radiology website (16) and maintain optimal accuracy in the diagnosis of malignancy, the MRI protocol should contain the minimum technical requirements listed in Table 1, including dynamic contrast-enhanced (DCE) imaging with time intensity curves (TIC) (16). When DCE with TIC are not available, a single 30–40-second T1-weighted



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Abbreviations: DCE = dynamic contrast-enhanced, DWI = diffusion-weighted imaging, FOV = field of view, IP = in-phase, OP = opposed-phase, O-RADS = Ovarian-Adnexal Reporting and Data System, PACS = picture archiving and communication system, PPV = positive predictive value, ROI = region of interest, 3D = three-dimensional, TIC = time intensity curve, T1WI = T1-weighted imaging

TEACHING POINTS

- The echo times and repetition times should be optimized to maintain high signal intensity in fluid-filled structures (bladder and spinal canal) and to optimize tissue contrast.
- The recommended higher *b* value is between 1000 and 1200 sec/mm² to ensure that fluid signal intensity is suppressed.
- TIC analysis is required to obtain the listed PPVs seen in the American College of Radiology O-RADS MRI risk score table.
- If DCE is not available, the nondynamic 30–40-second time point should be used for comparing enhancing adnexal lesion solid tissue to myometrium. It is important to note that at more delayed imaging, solid tissue may enhance less than the myometrium, even in O-RADS MRI 5 lesions.
- When using the nondynamic postcontrast analyses, the PPV and accuracy of malignancy diagnosis will be decreased.

gradient-echo series may be substituted (17). Required imaging sequences with specific technical parameters are needed to ensure that both the tissue contrast is optimal for differentiating the various fluid and solid component types and the spatial resolution is high enough to identify small enhancing nodules and papillary projections. This article reviews the proper O-RADS MRI scanning technique, how to optimize sequence parameters, and the most common pitfalls when the correct technique is not used for the assessment of adnexal lesions.

Patient Preparation

American College of Radiology does not have specific recommendations regarding patient preparation. However, there is evidence that some techniques may help in image quality (18). For example, asking the patient to avoid food for 3–6 hours before the examination can help slow bowel peristalsis, which can improve image quality. To further reduce bowel artifact, administration of an antiperistaltic agent, such as 1-mg intramuscular or intravenous injection of glucagon or 20-mg intramuscular injection of butylscopolamine bromide (Buscopan; Sanofi), can be considered, though is not required (18–20). A partially distended bladder helps reposition the uterus and adnexa toward the midpelvis, improving visualization of pelvic anatomy. To achieve optimal bladder distention, patients should be instructed to void approximately 30–60 minutes before being positioned in the MRI scanner (18). Although vaginal gel contrast material may be useful in evaluating other female pelvic pathologic conditions, there is no conclusive literature to support its use in the evaluation of adnexal lesions, given the typical location of adnexal lesions away from the vaginal vault. Its use will unnecessarily

increase cost and invasiveness of the examination. An intravenous line should be placed for the administration of gadolinium-based contrast material, a requirement for using the O-RADS MRI risk score. Lastly, coaching the patient so they breathe regularly and calmly, with shallow breaths and without use of their abdominal musculature during non-breath-hold sequences (eg, diffusion-weighted imaging [DWI]), can also help image quality (21).

Hardware Requirements

An optimized MRI protocol for evaluation of adnexal lesions includes minimum hardware and sequence requirements. Clinical MRI systems with standard field strengths of 1.5 or 3 T can both be used (5,14,22). MRI systems with field strength of less than 1.5 T, including open MRI, should not be used due to the suboptimal signal-to-noise ratio and lower spatial resolution, which may obscure important solid enhancing tissue such as papillary projections. In addition, lower field strengths require longer scan time, which may introduce more motion artifact and render DCE impossible. Equipment monitoring should also be performed according to American College of Radiology and American Association of Physicists in Medicine Technical Standards for Diagnostic Medical Physics Performance Monitoring of MRI Equipment (23).

A multichannel phased array body surface coil is recommended and is composed of multiple small coils that decrease noise and increase overall signal-to-noise ratio (24). The multichannel phased array coil also allows for parallel imaging, a tool to decrease acquisition time (24). The surface body coil should be comfortably placed over the patient's pelvis.

Protocol Overview

O-RADS MRI risk assessment requires a multiplanar multiparametric MRI protocol to accurately predict the risk of malignancy. The pertinent sequences for adnexal lesion evaluation include T2-weighted turbo or fast spin-echo imaging without fat saturation, T1-weighted in- and opposed-phase (IP and OP) gradient-echo or T1-weighted Dixon imaging, DWI, DCE three-dimensional (3D) T1-weighted imaging (T1WI), or pre- and postcontrast nondynamic 3D T1WI with fat saturation. If DCE 3D T1WI is performed, a postcontrast T1-weighted gradient-echo or T1-weighted Dixon sequence can still be useful to detect small solid tissue since DCE sequences may have lower spatial resolution to obtain the 15-second minimum for temporal resolution. Minimum parameters are listed in Table 1 and detailed scanning parameters are listed in Table 2.

The field-of-view (FOV) for each sequence should include the adnexal lesion in its entirety. Large lesions may require longer scan times or adjustments of section thickness and specific imaging planes to ensure reasonable acquisition time for patient comfort and image quality. However, small papillary projections may be missed with thicker sections (25). For example, when there is a lesion greater than 15 cm, the section thickness may need to be increased to 3.5 or 4 mm, and if the FOV is 28 cm, it may need to be increased to 30 or 32 cm to cover the entire lesion in the axial plane. It is also possible to change the scanning plane for the postcontrast images from axial to sagittal to achieve appropriate coverage and timing. Having two

Table 1: O-RADS MRI Recommended Protocol for Adnexal Lesions

Sequences	Plane*	Section Thickness*	Additional Requirements and Comments
T2WI FSE or TSE without fat saturation	Sagittal or coronal	4 mm	...
T2WI FSE or TSE without fat saturation	Axial	3 mm	...
T1WI IP and OP GRE or Dixon T1WI	Axial	3–4 mm	Can use the IP and OP images from 3D T1WI precontrast if using the Dixon technique, but these should be 3 mm
Diffusion-weighted echo planar imaging	Axial	4–5 mm	Low <i>b</i> values: 0 or 50 sec/mm ² High <i>b</i> value: 1000–1200 sec/mm ² If 4-mm section thickness, skip 2 mm; if 5-mm, skip 1 mm
DCE 3D T1WI GRE (pre- and postcontrast in same series)	Axial	3 mm	Pre- and postcontrast in single series are required to perform subtraction series Temporal resolution: ≤15 sec Continuous acquisition, starting with precontrast phase, injection of contrast material after precontrast phase is acquired, and continuing for total of 3 min
Nondynamic 3D T1WI GRE with fat saturation (pre- and postcontrast in separate series)	Axial	3 mm	Obtain one series precontrast If not acquiring a DCE MR image, obtain a minimum of one phase scanned at 30–40 sec after end of contrast material injection; additional time points can be obtained at 30-sec intervals, as desired If acquiring a DCE MR image, obtain a postcontrast series after DCE is completed with same parameters as precontrast 3D T1WI, for subtraction purposes

Source.—Adapted, with permission, from the American College of Radiology (69).

Note.—Recommended equipment: 1.5 or 3 T with a multichannel array body surface coil. FSE = fast spin echo, GRE = gradient echo, TSE = turbo spin echo.

* For very large lesion changes in plane and section thickness, adjustments may be needed to assure coverage of the entire lesion. However, small papillary projections may be missed.

series pre-prescribed with different FOVs and parameters may be a way to minimize the need for monitoring patient scanning and allowing the MRI technician to decide which series to use.

T2-weighted Imaging

Rationale and Benefits

T2-weighted imaging (T2WI) without fat saturation is advantageous for its high tissue contrast in the female pelvis given the many tissue types in a small space. It is especially important in determining the origin of a pelvic lesion (adnexal vs nonadnexal). Understanding the origin of a pelvic mass determines whether O-RADS can be applied and also allows for an accurate differential diagnosis. High-resolution T2WI is used to detect and characterize the morphology of solid tissue, which can be on the order of millimeters in size (eg, nodules, papillary projections). Detection of lymphadenopathy, peritoneal metastasis, and ascites can also be performed in the setting of high contrast between fat, fluid, and solid tissue (25,26). The T2WI sequence is fluid sensitive and is used in combination with the T1WI sequence to characterize the fluid in adnexal lesions (simple vs nonsimple) and to characterize solid components in an adnexal lesion (eg, fibroma vs bulk fat vs neoplastic solid tissue) (Fig 1) (26).

O-RADS MRI Recommendations

Non-fat-saturated T2WI is recommended for best tissue contrast in the pelvis. At least two T2WI planes should be used,

including a true axial plane and one orthogonal to it (sagittal or coronal). Imaging should be optimized to maximize resolution and to characterize solid and fluid components of lesions based on signal intensity. A small FOV (eg, 28 cm), with a large matrix (eg, 320 × 224) is recommended to maximize resolution and should include the uterus, ovaries, and adnexal lesion. Minimizing the FOV and maximizing the matrix size allows for the smallest voxel size, leading to maximizing resolution (27). At least the axial plane should also have a section thickness of 3 mm to ensure small papillary lesions are not missed between sections (Fig 2). The echo times and repetition times should be optimized to maintain high signal intensity in fluid-filled structures (bladder and spinal canal) and to optimize tissue contrast.

Tips and Tricks

To optimize T2WI, it is important to use fast spin-echo T2WI that uses multiple acquisitions with an echo train technique, acquiring a set number of sections (eg, fast-recovery T2 fast spin echo), compared with a T2WI sequence that uses a single acquisition to acquire all the sections (eg, single-shot T2 turbo spin echo). Since fluid has a long T1 relaxation time, a long repetition time, such as 3000 msec or greater, is needed to allow full recovery and reduce the T1 effects in the tissue. It is also important to choose an echo time that is long enough to maximize tissue contrast differences. Typically, an echo time between 80 and 110 msec allows for the difference in transverse magnetization to be optimal, while maintaining sufficient signal-to-noise ratio (28).

Table 2: Examples of Sequence Parameters

Sequence Plane	Coronal T2WI	Axial T2WI	Axial DWI	Pre-T1 Axial*	TIWI Dynamic	Post-T1WI Axial
Sequence type	2D FRFSE	2D FRFSE	2D EPI	3D Lava-Flex	3D DISCO	3D Lava-Flex
FOV (cm)	28	28	40	32	32	32
Section/skip [†]	4/0.5	2.9/0.3	5/1	3.2	3	3.2
TR [†] (msec)	3500	4500	10 500	6.1	5.9	6.1
TE [†] (msec)	85	85	95	3.1	2.9	3.1
TI [†]	...	...	94	...	...	...
Number of locations [†]	...	...	...	82	80	82
Number of phases	...	...	...	...	16	...
Scanning matrix	320 × 224	320 × 224	120 × 100	256 × 192	224 × 192	256 × 192
NSA	2	2	8	0.7	0.7	0.7
<i>b</i> value	...	...	50, 1200	...	...	...
Time [†]	2–4 min	2–4 min	4–5 min	20–25 sec	4 min	20–25 sec

Note.—Examination performed on a 1.5-T GE scanner. DISCO = differential subsampling with Cartesian ordering, EPI = echo planar imaging, FRFSE = fast recovery fast spin echo, NSA = number of signal acquired, TE = echo time, TI = inversion time, TR = repetition time.

* Using a Dixon-based sequence allows the IP and OP images to be reconstructed from this precontrast acquisition.

[†] These parameters and time will change depending on the prescribed volume based on vendor sequence specifications for acquisition.

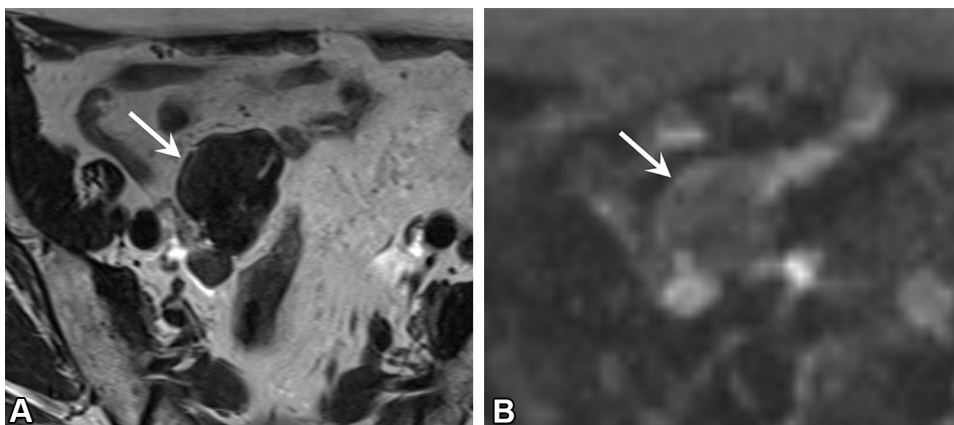


Figure 1. Fibroma proven with pathologic findings in a 78-year-old patient. **(A)** Axial T2-weighted image shows a right adnexal lesion (arrow), which is homogeneously T2 hypointense, similar to the iliopsoas muscle. **(B)** Axial high-*b*-value (*b* value = 1000 sec/mm²) diffusion-weighted image of the lesion (arrow) also shows low signal intensity. The lesion was later resected and confirmed to be a fibroma at histopathologic analysis.

One challenge in optimizing T2WI is overcoming motion-related artifact, which is common given the long acquisition time. Sources of motion artifact range from breathing, bowel peristalsis, uterine peristalsis, and patient movement. A motion-robust radial k-space acquisition sequence (eg, Siemens BLADE, Philips MultiVane, GE PROPELLER) mitigates most motion by oversampling central k-space representative of contrast data important for anatomy and can be performed as a repeated sequence if the conventional Cartesian T2WI is motion degraded on first pass (29). The cost is increased scan time of 1.2–1.5 times for similar resolution and coverage (30).

Breathing-related ghosting artifact is one of the most common at T2WI. To mitigate this artifact, saturation bands, which suppress signal, may be placed cranially over the upper abdomen and over the anterior abdominal wall, extending a centimeter anterior to the abdominal wall to account for motion during breathing (Fig 3). Changing the phase-encoding direction may also decrease breathing motion, since motion artifact is more prominent through the phase-encoding direction (Fig 3). There may be the cost of time for changing phase-encod-

ing direction as an increase in the number of phase-encoding steps increases acquisition time. As mentioned in the patient preparation section, bowel peristalsis may be minimized through patient preparation techniques before imaging.

Deep learning–based T2WI turbo spin echo or fast spin echo is now available commercially and may be a solution to decreasing acquisition time. Most deep learning–based T2-weighted images decrease the number of averages and undersample k-space in a manner similar to parallel imaging techniques to decrease acquisition time by up to threefold (31–33). A deep learning algorithm is then used to recuperate image quality. The advantage is that the training time is an upfront cost before imaging, which allows the postprocessing at the time of imaging to be quick. Studies have shown comparable image quality to conventional T2WI. However, there are some challenges in optimizing the sequences, especially in regards to motion artifact as the number of averages are usually decreased (31–34). The studies used to train the sequences are not institution specific, thus may not be optimized for an institution's patient population. Finally, the deep learning algorithm often decreases noise



Figure 2. Adnexal lesion containing small papillary projections in a 61-year-old patient. **(A)** Papillary projections (arrows) are not visualized on the axial T2-weighted image with 6-mm section thickness, a matrix of 256×260 , and an FOV of 40 cm with suboptimal spatial resolution. **(B)** The papillary projections (arrows) are visualized on the axial T2-weighted image with 3-mm section thickness, a matrix of 320×224 , and an FOV of 28 cm. **(C)** Axial diffusion-weighted image (b value = 1000 sec/mm²) confirms the papillary projections (arrows).

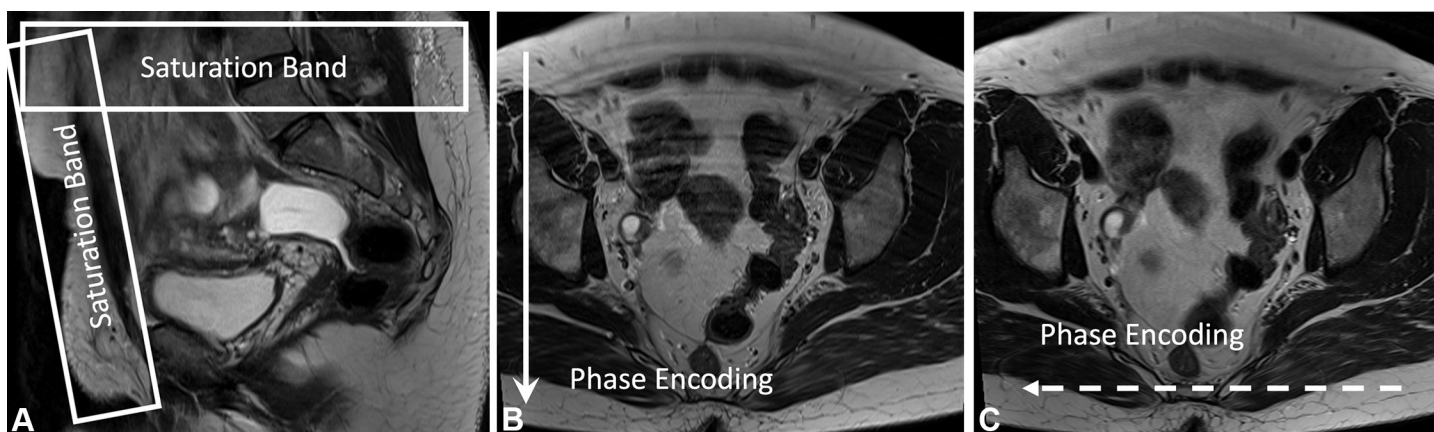


Figure 3. Correction of motion artifact at T2WI. **(A)** Sagittal T2-weighted image shows that saturation bands may be placed along the anterior abdominal wall and/or in the upper abdomen to mitigate breathing motion. **(B)** Axial T2-weighted image with phase encoding in the anterior-posterior direction (arrow) shows ghosting artifact caused by breathing, which appears as semicircular bands in the anterior abdomen. **(C)** Axial T2-weighted image shows that the ghosting artifact from breathing can be decreased through changing the phase-encoding direction from the anterior-posterior direction to the right-left direction (arrow).

to improve signal-to-noise ratio, which creates a different noise profile than one to which the reading radiologist is accustomed (31). Although deep learning T2WI is promising, and there have been a few studies that demonstrate its utility and quality, it is a new sequence and not widely adopted yet, thus there may be unforeseen issues that arise. Optimizing the sequence with the vendor for visualization of adnexal lesions is recommended before implementation.

T1-weighted IP and OP Imaging

Rationale and Benefits

IP and OP imaging is a gradient dual-echo T1WI sequence that capitalizes on the differences of resonance frequency between water and fat. In female pelvic imaging, it identifies fat in an adnexal lesion, which characterizes the lesion as benign in most cases (35). At the interfaces of water and fat, the voxels with similar amounts of water and fat will have signal suppressed, known as the India ink artifact (36,37). Bulk fat in ovarian dermoids can be identified on OP images through visualization of this artifact (Fig 4) (35). IP and OP imaging can also be used in conjunction with a fat-saturated sequence to distinguish between intravoxel fat content (eg, liquefied fat in dermoids)

and hemorrhagic material (Figs 5, 6). Bulk fat and hemorrhage both demonstrate high signal intensity on T1-weighted images without fat saturation (IP and OP), but hemorrhage remains with high signal intensity on fat-saturated T1-weighted images. Furthermore, IP imaging is particularly sensitive to susceptibility artifact and can detect hemosiderin, calcification, or surgical material from prior procedures. Finally, the sequence is fast, adding value with low cost.

O-RADS MRI Recommendations

The IP and OP sequence should be acquired in a true axial plane. FOV should include the uterus, the ovaries, and the entire adnexal lesion. Section thickness should be 3–4 mm.

Tips and Tricks

IP and OP imaging relies on the phase cycling of the resonant frequencies of the water and fat molecules, which is dependent on the field strength. OP is acquired at 2.2 msec (1.5 T) and 1.1 msec echo time (3 T), and IP is acquired at 4.4 msec (1.5 T) and 2.2 msec (3 T) echo time (38,39). To maximize the greatest signal differences between water and fat, each IP and OP image must be acquired at these specific TEs. As mentioned previously, subtle presence of liquid fat in adnexal lesions can

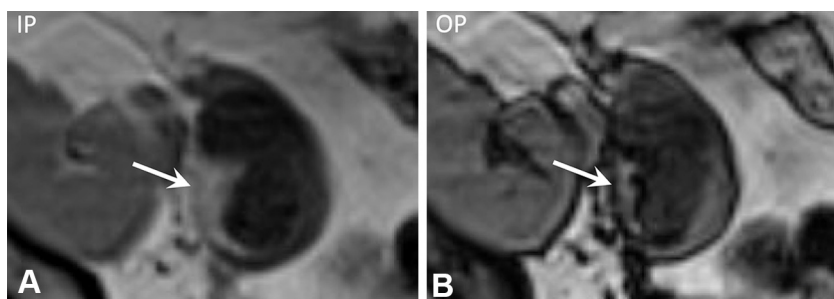


Figure 4. Dermoid on axial T1-weighted GRE images in a 37-year-old patient. **(A)** IP image shows higher signal intensity in a right adnexal lesion (arrow) when compared with the OP image. **(B)** OP image shows lower signal intensity in a right adnexal lesion (arrow) compared with the IP image, compatible with liquefied fat in a dermoid, which can manifest as liquefied fat.

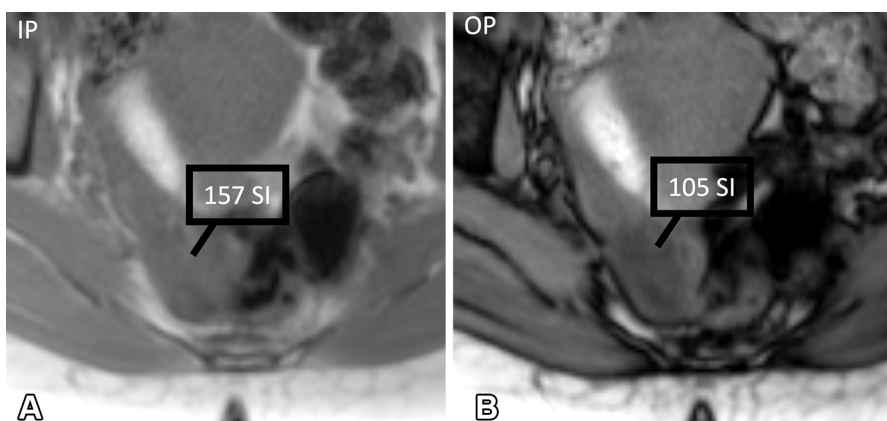


Figure 5. Dermoid on axial T1-weighted GRE images in a 27-year-old patient. **(A)** IP image shows higher signal intensity (157 SI) in a right adnexal lesion. **(B)** OP image shows lower signal intensity (105 SI) in a right adnexal lesion compared with the IP image, compatible with liquefied fat in a dermoid.

differentiate a benign dermoid from lesions with more malignant potential containing hemorrhage or protein (38,39).

Signal is lost over time, thus OP images should be acquired before IP images so that signal decay does not obscure signal loss from fat and water interaction (38). The IP and OP images can be acquired separately or can be part of the 3D Dixon-type technique. The 3D Dixon-type technique capitalizes on chemical shift properties and acquires two sets of spin echo sequences at echo times slightly different from each other, again utilizing the difference in resonance frequencies of water and fat (40). The technique allows various combinations of water and fat signals in the image reconstructions (40). Similar to IP imaging, water and fat signals can be combined together, and similar to OP imaging, fat signal can be subtracted from water signal. In addition, water-only and fat-only signal images can be reconstructed. The water-only sequence can be used as T1WI with fat saturation for nondynamic pre- and postcontrast imaging, as described in subsequent sections (41). The Dixon technique does require more acquisition time compared with IP and OP imaging. However, one acquisition can produce multiple sequences in the adnexal lesion protocol, decreasing overall scan time. One disadvantage of the Dixon technique is the fat-water swap error, which often occurs at field inhomogeneities or in regions of complex tissue composition (41,42).

Diffusion-weighted Imaging

Rationale and Benefits

DWI, performed using an echo-planar sequence, is a functional technique that assesses water molecule diffusivity. Restricted diffusion, as seen in hypercellular malignant tumors, results in high signal intensity at high b values. The O-RADS

MRI system leverages this property to differentiate benign from malignant solid tissue. DWI is particularly useful for detecting benign fibrous tissue, which appears hypointense at both high b -value DWI and T2WI due to the high collagen content and lack of water molecules in fibrous tissue (43). DWI can highlight lymphadenopathy and small papillary projections and peritoneal nodules, which stand out as areas of abnormal high signal intensity against the suppressed background of the remainder of the adnexal lesion. DWI is also important to use in conjunction with postcontrast imaging as abscesses and hemorrhagic and/or proteinaceous material may restrict diffusion but are benign processes.

O-RADS MRI Recommendations

O-RADS MRI recommends that DWI be acquired in a true axial plane. Section thickness of 4–5 mm should be used. If 4-mm section thickness is used, then a 2-mm section gap is recommended to prevent cross talk between two adjacent sections (44,45). If 5-mm section thickness is used, then a 1-mm section gap is recommended. At least two b values are recommended, with the lower b value between 0 and 50 sec/mm^2 , with preference to 50 sec/mm^2 to avoid confounding signal in the adjacent vascular structures (46). The recommended higher b value is between 1000 and 1200 sec/mm^2 to ensure that fluid signal intensity is suppressed (Fig 7) (25).

Tips and Tricks

Susceptibility artifact is a major challenge in optimizing DWI. The artifact is usually worse in the phase-encoding direction, which is usually in the anteroposterior direction since it is typically the shortest distance and requires the least number of phase-encoding steps. If the offending object is in line with the adnexal lesion, changing the phase-encoding direction

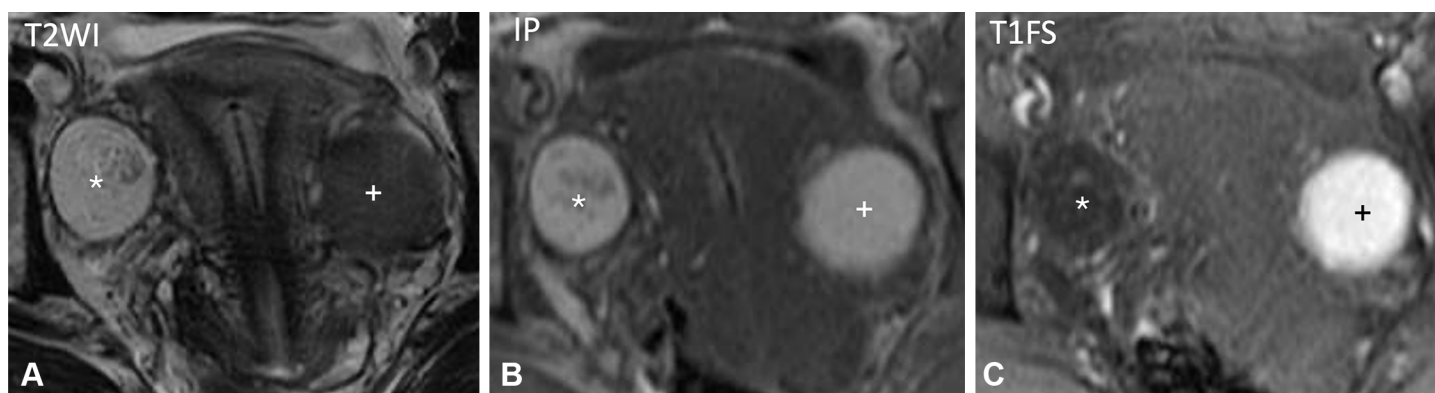


Figure 6. Right adnexal dermoid and left adnexal hemorrhagic cyst in a 32-year-old patient. **(A)** Axial T2-weighted image shows high signal intensity in the right adnexal dermoid (*) and low signal intensity in the left hemorrhagic cyst (+), as fat appears bright at T2WI and hemorrhagic material appears dark. **(B)** Axial IP image shows high signal intensity in the right adnexal dermoid (*) and left adnexal lesion (+) when compared with those on the OP image. **(C)** Axial T1-weighted fat-saturated (*T1FS*) image shows low signal intensity in a right adnexal lesion (*) compared with the IP image, compatible with saturated bulk fat in a dermoid, whereas the left adnexal lesion (+) maintains high signal intensity from the hemorrhagic cyst.

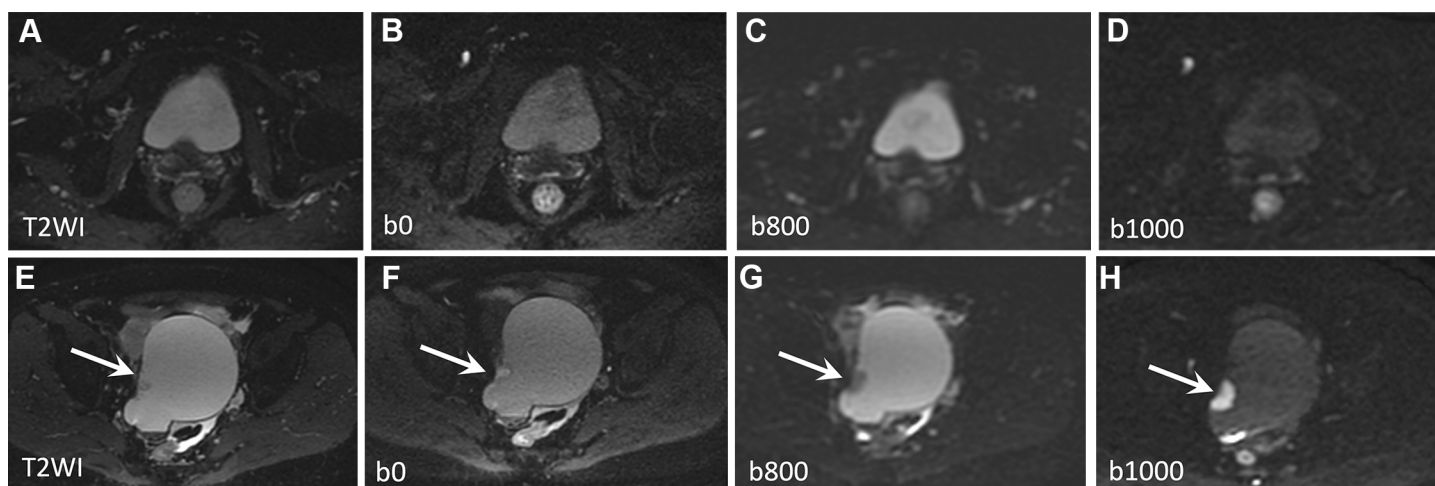


Figure 7. High b value at axial DWI increases conspicuity of solid tissue. **(A–D)** Titrating the correct b value is achieved by evaluating the bladder on the axial T2-weighted image **(A)** and on images with progressive b values **(B–D)** to choose a high b value that suppresses fluid signal intensity, for example, at a b value of 1000 sec/mm^2 **(D)**. **(E–H)** The solid tissue (arrow) in an adnexal cyst is difficult to see on the axial T2-weighted image **(E)**, demonstrates increasing conspicuity on the b -value 0 sec/mm^2 **(F)** and b -value 800 sec/mm^2 **(G)** images, and is most clearly seen on the b -value 1000 sec/mm^2 image **(H)** because the background fluid signal intensity of the cyst is suppressed at high b values.

may mitigate the artifact (eg, avoiding the susceptibility artifact from a total hip arthroplasty). A small FOV DWI sequence, which essentially places saturation bands on the left and right periphery, may decrease susceptibility artifact from hip implants, as the saturation bands suppress signal (47). The amount of susceptibility artifact is proportional to the strength of the magnetic field. Thus, the 3-T scanner system produces more susceptibility artifact when compared with the 1.5-T system, so switching to a lower Tesla scanner system may also decrease artifact (48). If bowel gas is persistently a problem, patient preparation with micro enema may be attempted.

Interpreting DWI can be challenging, as high signal intensity on high b -value images is not specific to malignancy. Differential diagnoses include nonneoplastic entities, such as tubo-ovarian abscesses or cysts containing hemorrhagic or proteinaceous material or keratin in dermoids. Correlation of high DWI signal intensity with enhancement ensures the signal is from soft tissue and not diffusion-restricting fluid

or other nonenhancing solid components (49). In the current O-RADS MRI system, the role of DWI remains limited—primarily supporting the O-RADS MRI 2 and serving as an adjunct in assessing diffusion restriction in solid tissues suspicious for malignancy, rather than being used independently to assign higher scores. The assessment of the apparent diffusion coefficient map is also not specified in the system. However, it may be helpful for identifying true diffusion restriction and excluding T2 shine-through seen in benign lesions and has been reported to aid in identifying lesions that correspond to higher O-RADS MRI scores (49–51).

Dynamic Contrast-enhanced Imaging

Rationale and Benefits

The dynamic TIWI series is a spoiled 3D gradient-echo series with or without fat saturation acquisition optimized for temporal resolution, with a minimum of a 15-second time resolution

using a 3-mm section thickness. This series is acquired with phases before contrast material administration, then while contrast material is injected for up to 3 minutes. TIC analysis is then performed on the picture archiving and communication systems (PACS) workstation or off-line. TIC analysis is required to obtain the listed PPVs seen in the American College of Radiology O-RADS MRI risk score table (17).

DCE is critical for accurate cancer risk prediction and detection of early enhancement seen in malignant neoplasms (17,52,53). Malignant ovarian neoplasms require angiogenesis, the development of a neovascular blood supply, to grow and metastasize (54). Although there is angiogenesis in normal ovarian tissue, there is also regulated regression of vasculature and follicular atresia during the normal ovarian cycle. In tumoral growth, there is improper regulation and vascular growth is uninhibited. The unfettered angiogenesis in ovarian cancer with a low pericyte coverage index is manifested at imaging by rapid early enhancement (53,55). With high temporal resolution, DCE can capture the early enhancement of angiogenesis, and high risk curves predict potential malignancy with a PPV of greater than 90% (17,56).

O-RADS MRI Recommendations

First, the axial plane should be obtained, and the FOV should include the entire adnexal lesion. Other planes may be chosen if the lesion coverage would be more ideal (25). Consideration of the DCE analysis software is important. Some vendors do not allow plotting of regions of interest from different sections. In such a circumstance, axial plane may be optimal since the uterus and adnexa are more likely to be included in the same image. The sequence should have a section thickness of 3 mm or less to characterize potential small papillary projections and the temporal resolution should be at least every 15 seconds. To ensure capture of early arterial enhancement, the acquisition should be started slightly before the start of the contrast material injection. Contrast material should be administered intravenously at a 2 mL/sec rate to ensure an adequate bolus, and the sequence should be at least 3 minutes in total (5,22).

Tips and Tricks

Dynamic 3D T1WI sequences with fat saturation are fast sequences. To achieve the short acquisition time needed for the O-RADS MRI–stipulated time-resolution of 15 seconds, several techniques can be used. Parallel imaging may be used to accelerate images through undersampling of k-space with minimal, if any, loss of image quality (57). Golden-angle radial MRI techniques are motion robust and produce high-resolution imaging with the ability to reconstruct at any time resolution (eg, Siemens Healthineers GRASP, Philips eTHRIVE) (58). Other sequences normally used for MR angiography may also be explored (eg, Siemens Healthineers TWIST, GE Healthcare TRICKS, Philips 4D-TRAK). However, there is a cost of decreased spatial resolution in these sequences (59). Advanced deep learning–based accelerating techniques are currently in development and will likely be available in the short term (60,61).

For large lesions, adjustments in the FOV and section thickness can help in achieving the appropriate time resolution. For example, changing the FOV from 32 cm to 34 or 36 cm and

the section thickness from 3.0 to 3.2 mm would decrease the time it takes for each phase to be acquired. The partial FOV and the matrix can also be adjusted to obtain a temporal resolution of 15 seconds. Lastly, changing the scanning plane from axial to sagittal can also decrease scan time in some patients.

Nondynamic Postcontrast 3D T1WI

The nondynamic T1WI series is a spoiled 3D gradient-echo with fat saturation acquisition optimized for higher resolution, which is particularly helpful for detecting small papillary projections at postcontrast imaging (62,63). The short acquisition time allows for the sequence to be performed within a single breath hold, which minimizes motion artifact (62). The nondynamic T1WI series is recommended regardless of acquiring DCE, as the nondynamic T1WI series have higher resolution and can be performed after the DCE.

Precontrast 3D T1WI with Fat Saturation

Rationale and Benefits.—T1WI with fat saturation is performed without intravenous contrast material to identify bulk fat and presence of hemorrhage or proteinaceous material in adnexal lesions and is most useful in conjunction with IP and OP imaging (Fig 6) (64). With the fat saturation, bulk fat has low signal intensity and hemorrhagic and proteinaceous material have high signal intensity (Fig 8) (64). Additionally, it is important to have a precontrast series with identical imaging parameters to those of the postcontrast series so that subtraction can be performed to detect enhancement of solid tissue with intrinsic hyperintense T1 signal (Fig 9) (26).

O-RADS MRI Recommendations.—The precontrast T1-weighted images should be obtained in a true axial plane. However, if the lesion is more clearly seen in the sagittal or coronal view, these planes can be used. Another option is to use an axial 3D T1WI sequence that makes coronal and sagittal reformatting feasible. A small FOV should be used (eg, 32 cm) and centered on the uterus and ovaries, including all portions of the adnexal lesion, with a section thickness of 3 mm (25). However, for large lesions, the FOV and the section thickness maybe increase, for example, to an FOV of 34 or 36 cm and a section thickness of 3.5 or 4 mm. As previously mentioned, 3D T1-weighted fat suppressed images can be obtained using the Dixon technique for increased efficiency (63).

Tips and Tricks.—Incomplete and nonuniform fat saturation are common problems that may arise when performing T1WI (63). T1WI fat saturation is performed with a frequency-selective saturation pulse (65). Since the fat suppression frequency is predetermined, variations in the B_0 magnetic field or patient-specific factors can lead to suboptimal fat saturation. Field inhomogeneities may result in incomplete or uneven fat suppression. Performing shimming to improve B_0 homogeneity can help minimize these artifacts (63,65). Additional preparation radiofrequency pulses may be performed to improve fat saturation (63). Susceptibility artifact, resulting from disruptions in the field at soft-tissue or metal and air interfaces, causes nonuniform fat saturation. Using a lower field

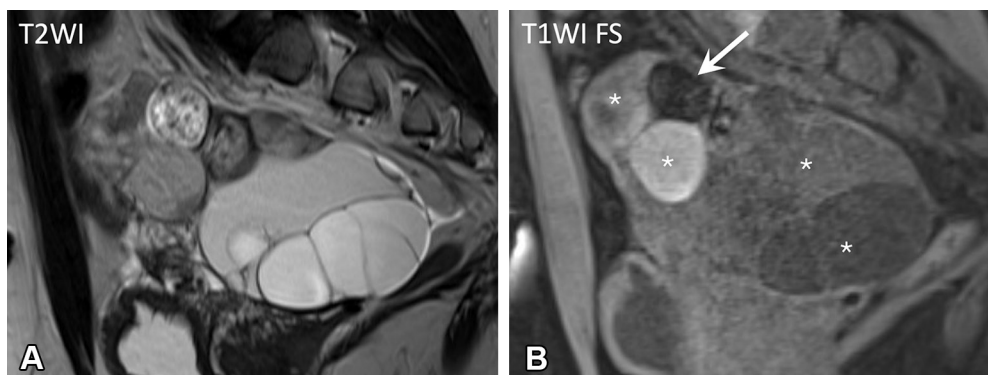
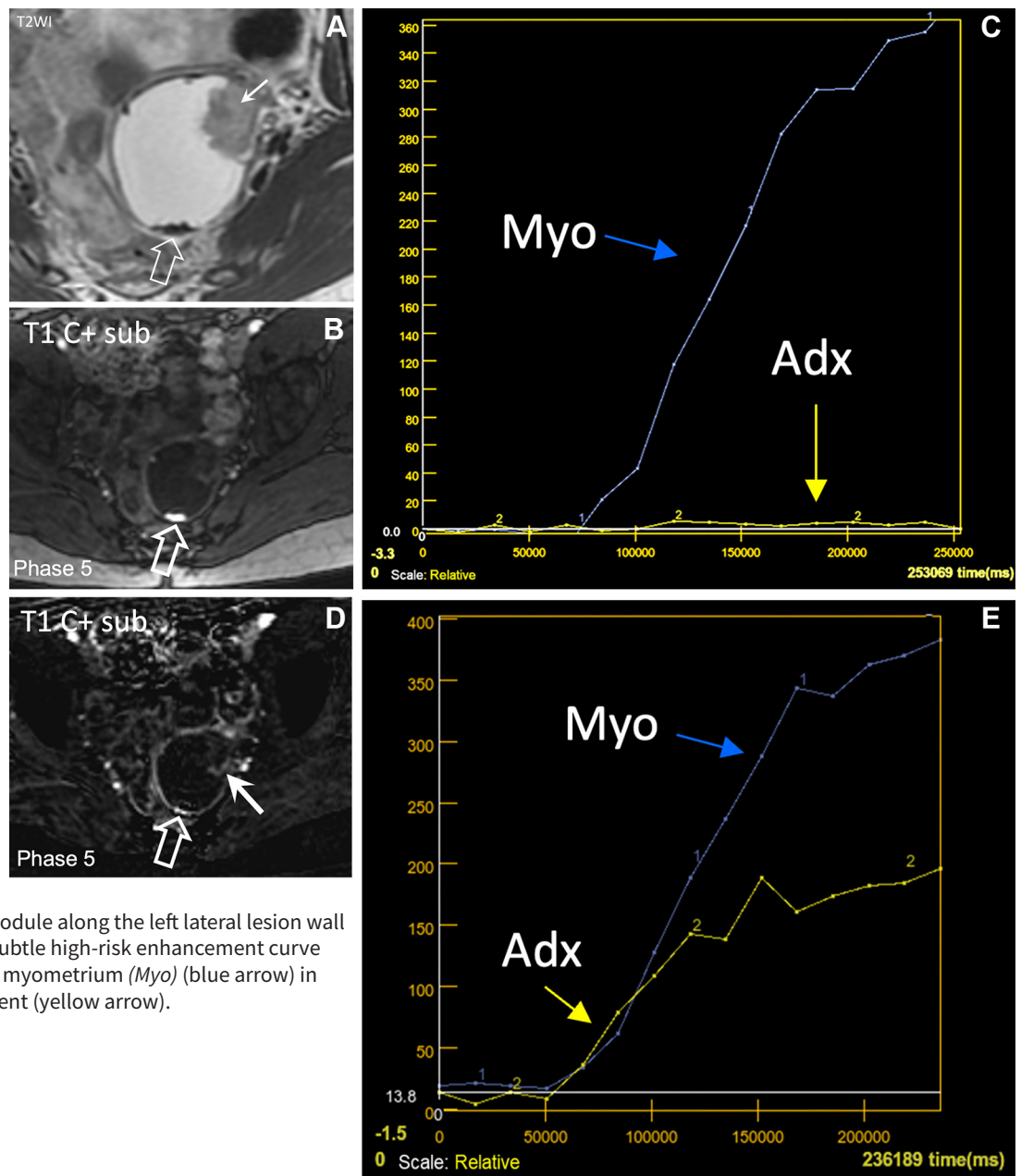


Figure 8. Mucinous cystadenoma in a 26-year-old patient with a multiloculated cyst. The cyst was later shown to be a mucinous cystadenoma. **(A)** Sagittal T2-weighted image shows a multiloculated cyst containing various shades of T2 signal intensity. **(B)** The multiloculated cyst contains locules (*) of various shades of T1 hyperintensity on the sagittal fat-saturated T1-weighted image, correlating with variable hemorrhagic and proteinaceous material. Susceptibility artifact is in a locale containing hemosiderin (arrow). FS = fat saturated.

Figure 9. Endometrioid adenocarcinoma in a 67-year-old patient. Subtraction images are important to identify enhancing solid tissue. **(A)** Axial T2-weighted image shows two solid components (arrows) in a left adnexal lesion. **(B)** Axial postcontrast T1-weighted image (T1 C+) shows nodular T1 high signal intensity along the posterior lesion wall that appears to enhance (open arrow). **(C)** On the associated TIC graph, generated on the Advantage Workstation (GE Healthcare), the adnexal curve (Adx) does not increase in signal intensity when compared to the myometrium curve (Myo) (blue arrow) over time and therefore does not enhance (yellow arrow).

(D) The same nodular T1 high signal intensity (open arrow) along the posterior wall does not enhance on the subtracted axial postcontrast T1-weighted image (T1 C+ sub) and represents an area of hemorrhage. This is hypointense on the T2-weighted image (open arrow in A). This is in contradistinction to the linear nodularity along the posterior aspect of the left lateral soft-tissue nodule (solid arrow). **(E)** This region of a nodule along the left lateral lesion wall is truly enhancing and displays a subtle high-risk enhancement curve (Adx) with a slope greater than the myometrium (Myo) (blue arrow) in the first few seconds of enhancement (yellow arrow).



magnet in a patient with a metallic orthopedic implant may improve fat saturation. Placing the patient in the center of the magnet and far from the edges of the bore may also improve image quality (27). Finally, using a Dixon technique may further improve fat saturation as the technique is more robust against field inhomogeneities through reliance on the chemical shift properties of water and fat (41).

Postcontrast 3D T1WI with Fat Saturation

Rationale and Benefits.—If DCE is not available, the nondynamic 30–40-second time point should be used for comparing enhancing adnexal lesion solid tissue to myometrium. It is important to note that at more delayed imaging, solid tissue may enhance less than the myometrium, even in O-RADS MRI 5 lesions. When using the nondynamic postcontrast analyses, the PPV and accuracy of malignancy diagnosis will be decreased (17,25,26).

O-RADS MRI Recommendations.—O-RADS MRI recommends that if DCE cannot be performed, postcontrast T1WI is performed 30–40 seconds after contrast material injection (25). This sequence should be a 3D gradient-echo sequence with fat saturation, high resolution, and a section thickness of 3 mm. Additional images after the 30–40-second series may be obtained but are not required.

Tips and Tricks.—Postcontrast T1WI should be performed with the same parameters as the precontrast T1WI examination to obtain a true subtraction image. Clear breathing instructions, especially in the patient's native language, can optimize registration between the two (21). End-expiration breath hold has been shown to decrease motion artifact (66,67). To further optimize registration, some PACS may allow manual fine-tuning of registration and may provide their own subtraction reconstructions.

Postcontrast Interpretation

O-RADS MRI scoring requires contrast-enhanced images when non-T2-weighted dark or DWI dark solid tissue is present. This assessment is most accurately performed using a dynamically acquired 3D T1WI gradient-echo series with TIC analysis. As noted previously, a nondynamic 3D T1WI gradient-echo series acquired at 30–40 seconds with visual inspection of the enhancement may be used. However, the PPVs will be less accurate for the risk score categories (14,17,26). For example, because the nondynamic technique does not allow TIC analysis, all of the lesions with low-risk curves cannot be scored as O-RADS MRI 3 and will be grouped into O-RADS MRI 4, decreasing the overall PPV of the score 4 category.

TICs can be generated using software equipped with a mean curve analysis function, either directly within the PACS environment or on an external workstation. Certain PACS vendors have TIC curve analysis functions available (eg, Sectra, Visage Imaging) whereas others do not. However, at sites where the PACS vendor does not offer TIC curve analysis within system, the dynamic series can be analyzed by postprocessing software, such as DynaCAD (Invivo), which is vendor

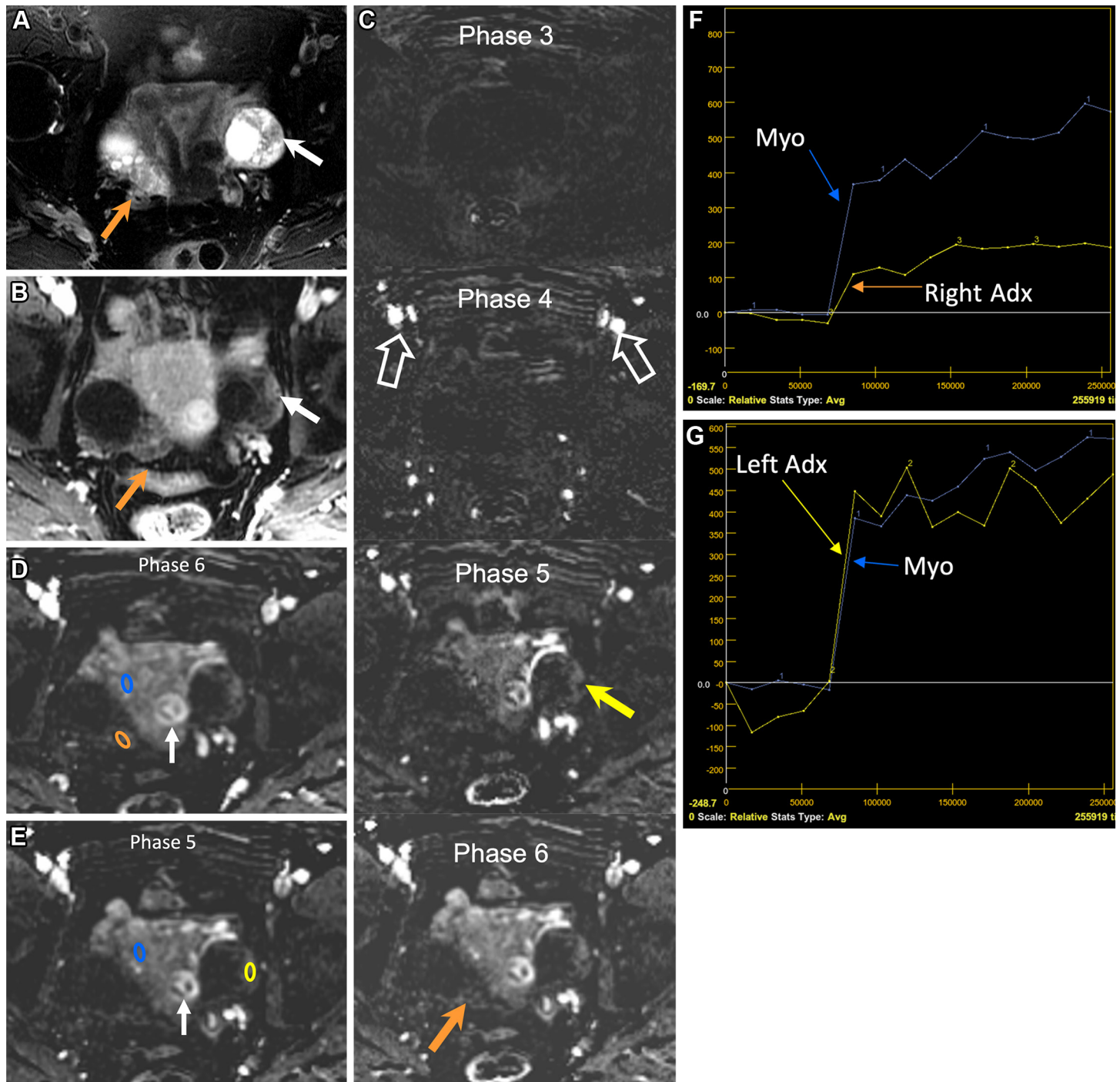
neutral, or Syngo Via for Siemens Healthineers scanners and the Advantage Workstation for GE Healthcare scanners. The process is similar to that used for dynamic assessments of the breast, prostate, or cerebral perfusion. Step 1 is to place a region of interest (ROI) in the outer myometrium of the uterus, avoiding fibroids and vessels on either the first or second phase in which the uterus enhances (Fig 10). Step 2 is to place a second ROI in the region of greatest enhancement of the solid tissue in the lesion, again utilizing the first or second phase in which the solid tissue is seen to enhance, taking care to place the ROI in the area of greatest enhancement. Once the TIC has been constructed, assess the graph to ensure the scale is set to “relative” or “percent” rather than “absolute,” as both curves may not be correctly displayed if “absolute” is used (Fig 11). Although the acquired images are used for ROI placement, it is important that the subtracted images are crosschecked to identify areas of enhancement, particularly when there are T1-hyperintense foci in a lesion. This will ensure that an area of T1-hyperintense signal is not mistaken for an area of enhancement when there is none (Fig 9).

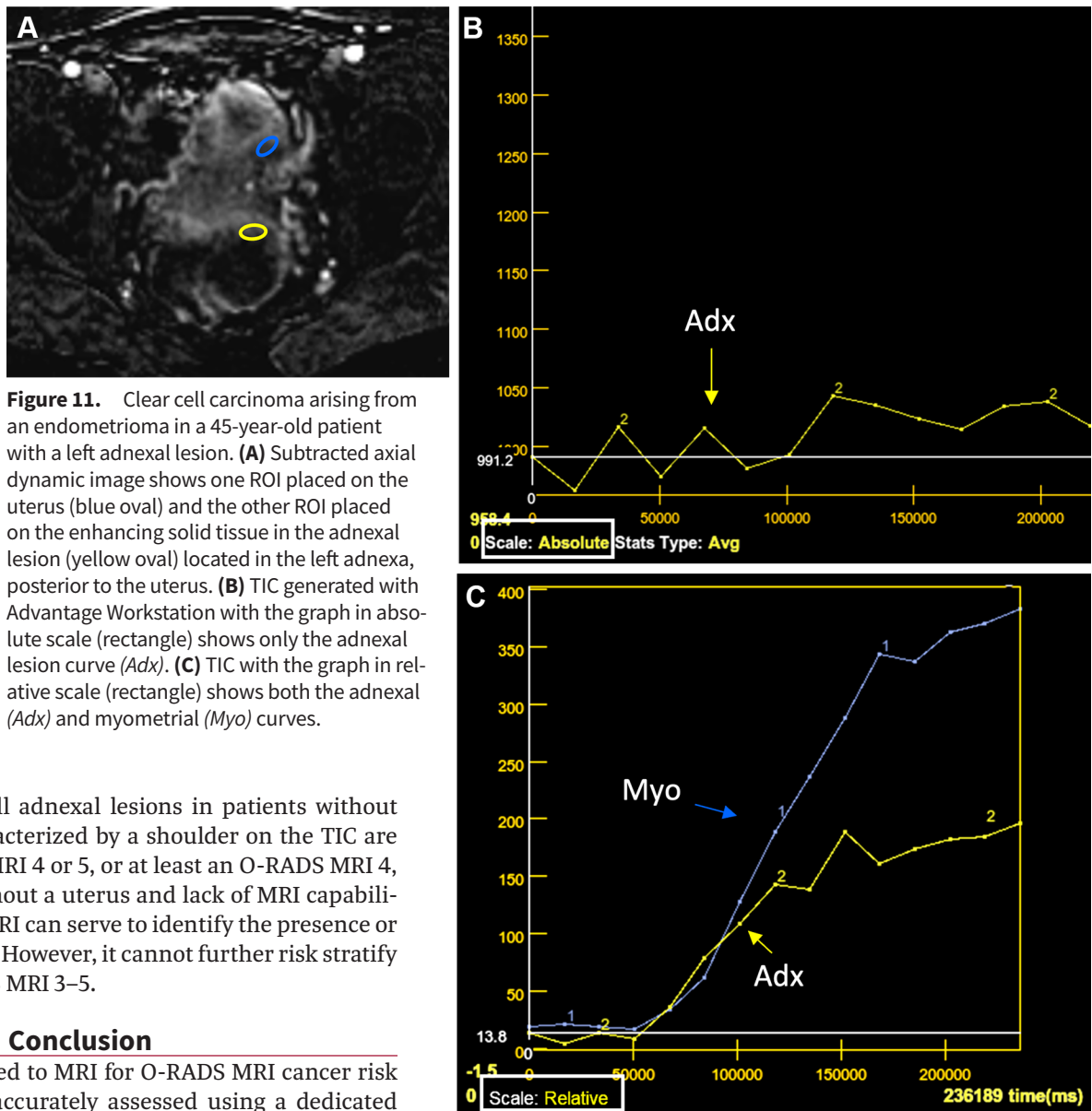
Once these troubleshooting steps are completed, the TIC curve of the solid tissue can be analyzed. This is accomplished by assessing the slope of the initial rise of contrast enhancement in the solid tissue versus the slope of the myometrial enhancement to assign the curve to a risk category as follows (Fig 12): (a) low-risk curve, minimal and gradual increase in signal over time with no well-defined shoulder and no plateau; (b) intermediate-risk curve, an initial slope less than or equal to the myometrium with a shoulder with a plateau; and (c) high-risk curve, an initial slope greater than the myometrium with a shoulder with a plateau. A low-risk curve classifies the adnexal lesion as O-RADS MRI 3, an intermediate-risk curve classifies as O-RADS MRI 4, and a high-risk curve classifies as O-RADS MRI 5 (25).

Without DCE, risk stratification of adnexal lesions is less accurate and lesions with enhancing solid tissue are classified as either O-RADS MRI 4, intermediate-risk lesion, or O-RADS MRI 5, high-risk lesion (17,68). Risk stratification between O-RADS MRI 4 and 5 is based on visual comparison of enhancement between the ROI of greatest enhancement in the adnexal lesion and the outer myometrium. If the solid tissue enhances less than or equal to the outer myometrium at the 30–40-second time point, then the lesion is classified as O-RADS MRI 4. If the solid tissue enhances more than the outer myometrium, then the lesion is classified as O-RADS MRI 5. Similar to the analysis of the DCE series, if the adnexal lesion has a T1-hyperintense area, utilization of subtraction images is needed to identify any underlying enhancing components. It is also essential that only the 30–40-second time point is used for this analysis, as on more delayed imaging, the solid tissue may enhance less than the myometrium, even in an O-RADS MRI 5 lesion.

In the absence of a uterus, DCE can still assist in risk stratification, though in a modified fashion. Without a uterus, TICs are classified into two categories instead of three: a low-risk curve (without a shoulder) or an intermediate- or high-risk curve (with a shoulder). Intermediate- and high-risk curves cannot be differentiated without comparison with the

Figure 10. Bilateral high grade serous carcinoma in a 45-year-old patient. **(A)** Axial T2-weighted image shows bilateral adnexal lesions with papillary projections and nodular areas along the wall (arrows). **(B)** Axial T1-weighted postcontrast image shows enhancement of the papillary projection and nodular areas (arrows). **(C)** Phases 3–6 of the subtracted axial dynamic images with temporal resolution of 15 seconds show no contrast enhancement in phase 3, contrast enhancement within the arteries in phase 4 (open arrows), enhancement of the nodule in the left adnexal lesion and uterus in phase 5 (yellow arrow), and subtle enhancement of the papillary projection in phase 6 (orange arrow). The subtle enhancement of papillary projections is often more clearly appreciated on nondynamic T1-weighted postcontrast images, as in this example, due to the higher resolution. **(D)** Phase 6 of the axial dynamic images shows ROI placement on the papillary projection (orange oval) in the center of the structure where there is the greatest area of enhancement. Another ROI is placed in the myometrium (blue oval), avoiding the fibroid (arrow). **(E)** Phase 5 of the axial dynamic images shows ROI placement on the nodule (yellow oval). Another ROI is placed in the myometrium (blue oval), avoiding the fibroid (arrow). **(F)** Analysis of the TIC generated on an Advantage Workstation shows that the right adnexal (*Adx*) slope is less than that of the myometrium (*Myo*) and is therefore an intermediate-risk curve. **(G)** Analysis of the TIC shows that the left adnexal (*Adx*) slope is greater than that of the myometrium (*Myo*) and is therefore a high-risk curve.





myometrium. Thus, all adnexal lesions in patients without a uterus that are characterized by a shoulder on the TIC are classified as O-RADS MRI 4 or 5, or at least an O-RADS MRI 4, lesion. In patients without a uterus and lack of MRI capabilities to perform DCE, MRI can serve to identify the presence or absence of solid tissue. However, it cannot further risk stratify the lesion into O-RADS MRI 3–5.

Conclusion

Adnexal lesions referred to MRI for O-RADS MRI cancer risk scoring can be most accurately assessed using a dedicated MRI protocol with optimized technical requirements. Use of a correctly optimized O-RADS MRI technique can decrease the number of benign lesions referred to gynecologic oncologists and surgery, while expediting the triage of patients with potential malignancy to cancer specialists.

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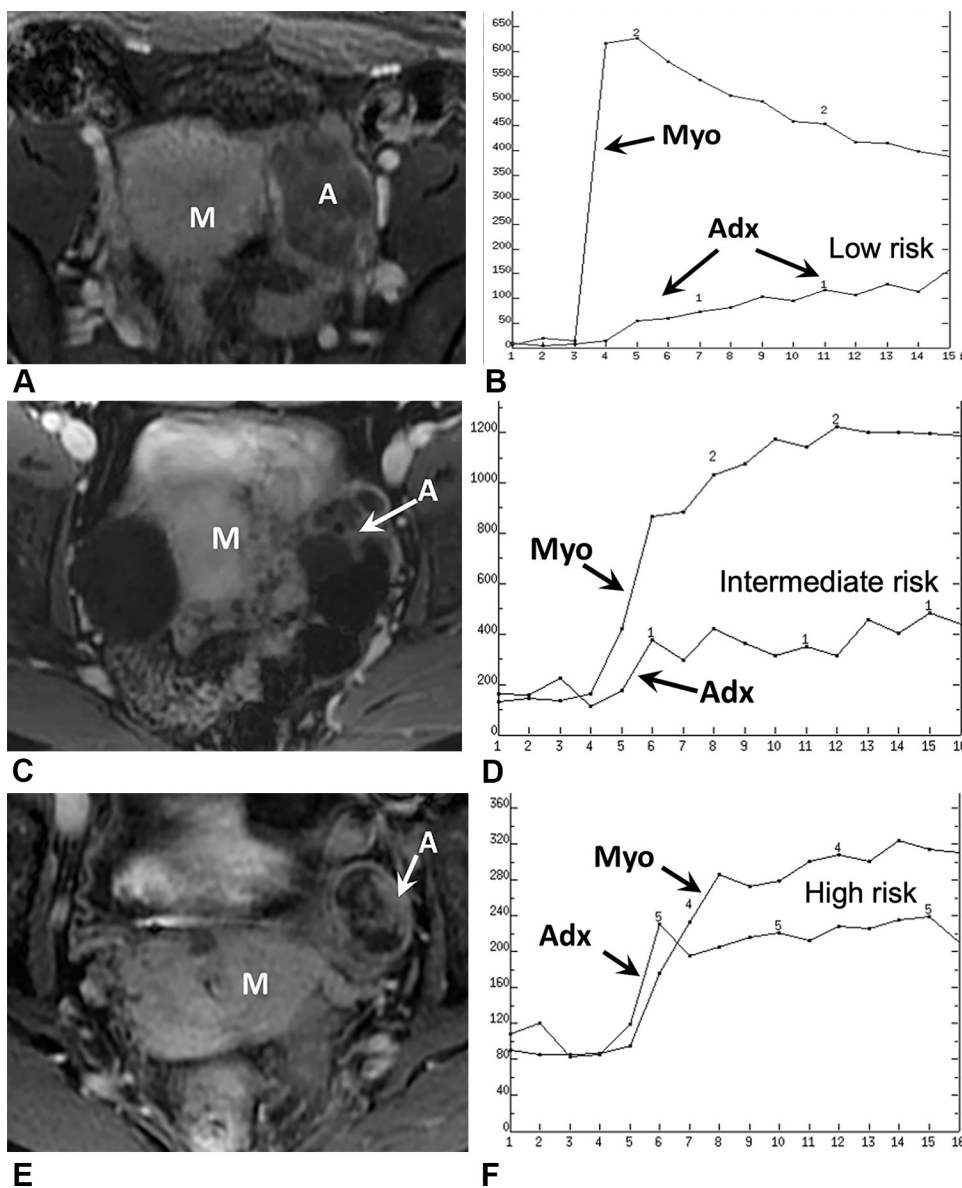


Figure 12. Three types of TIC characterizing adnexal lesions in three patients. **(A)** Enhancing solid tissue in a left adnexal lesion (A) on the axial T1-weighted contrast-enhanced image shows a low-risk curve. M = myometrium. **(B)** A low-risk curve is characterized by a slow gradual increase in signal over time with no well-defined shoulder and plateau (Adx) when compared with the myometrium (Myo). **(C)** Axial T1-weighted contrast-enhanced image of the enhancing solid tissue in the left adnexal lesion (A) shows an intermediate-risk curve. M = myometrium. **(D)** An intermediate-risk curve is characterized by a slope of enhancement (Adx) less than the myometrium (Myo) but with the presence of a shoulder and plateau. **(E)** Axial T1-weighted contrast-enhanced image of the enhancing solid tissue in the left adnexal lesion (A) shows a high-risk curve. M = myometrium. **(F)** A high-risk curve is characterized with slope of enhancement (Adx) greater than myometrium (Myo), and a well-defined shoulder and plateau.

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